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A natural polymer-based hydrogel with shape controllability and high toughness and its application to efficient osteochondral regeneration†

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Hydrogels prepared from sustainable natural polymers have broad prospects in the biological field. However, their poor mechanical properties and challenges in achieving shape control have limited their application. Herein, a novel preforming dual-effect post-enhancing method is proposed to address these issues. The method utilizes the hydrogen bonding of agar to obtain a shape-controllable preformed hydrogel at low polymer concentrations using casting, injection, or 3D printing techniques. Subsequently, the preformed hydrogel was subjected to a permeation process to form a post-enhanced multi-network (PEMN) hydrogel with hierarchical chain entanglements to ensure its high toughness, which exhibits tensile and compressive strengths of up to 0.51 MPa and 1.26 MPa with solely physically crosslinking networks. The excellent biocompatibility of the PEMN hydrogel prepared without the need for additional initiator agents under mild conditions was confirmed by both *in vitro* and *in vivo* experiments. Furthermore, the adaptability for irregular defects, suitable toughness, adhesive properties, and degradability of PEMN hydrogels are beneficial to provide mechanical support, induce endogenous cell mineralization, and accelerate the regeneration of cartilage and subchondral bone with more than 40% bone regeneration in 12 weeks. Our work has provided a novel solution to simultaneously achieve shape controllability and high toughness based on natural polymers among the already well-explored strategies for osteochondral regeneration.

New concepts

A novel sustainable natural polymer-based hydrogel with simultaneous shape controllability and high toughness is obtained by the proposed preforming dual-effect post-enhancing method entirely free from toxic chemical crosslinkers. The preformed hydrogel based on the hydrogen bonding of agar provides the adaptability for irregularly shaped defects, while the preformed hydrogel is subjected to a permeation process to form a post-enhanced hydrogel. Its multiple physically crosslinked networks give strong mechanical strength owing to the hierarchical chain entanglement. The biocompatibility, shape controllability, and suitable toughness are proven to induce endogenous cell mineralization and chondrocyte differentiation and promote the regeneration of cartilage and subchondral bone efficiently. This strategy provides a common way to prepare a kind of tough physically crosslinked hydrogel with outstanding translational prospects as the osteochondral repair scaffolds.

Introduction

In recent years, hydrogels have emerged as a promising material in the field of biomedical materials.^{1–3} The structural similarity of the hydrogels composed of 3D hydrophilic networks to tissues, as well as their high-water contents of 80–99% and low coefficient frictions, make them ideal alternative scaffolds for tissue engineering.^{4,5} An ideal hydrogel scaffold should possess good biosafety, strong mechanical support, and customized external geometry.^{2,6} With the emphasis on environmentally friendly concepts and the increasing cost of synthetic petroleum-based chemical polymers, sustainable natural polymer-based hydrogels derived from biomass have been developed.^{4,7–9} Also, their excellent biocompatibility, biodegradability, and tunable physicochemical properties position them as potential scaffolds for tissue engineering applications.

Although natural polymer-based hydrogels have received extensive attention in tissue engineering due to their abilities to infiltrate and regenerate the surrounding tissues, the contradiction between their mechanical support and molding controllability makes the preparation of the hydrogels with outstanding tissue regeneration effects challenging.^{6,10,11} In general, cartilage

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withstands the compressive stress of several MPa and the nominal compressive modulus of 0.1–1.0 MPa during human movement.¹² In practical application, conventional scaffolds based on natural polymer-based hydrogels usually show insufficient mechanical strength and endurance under cyclic loading and abrasion conditions.

Great efforts have been made to enhance the mechanical support of natural polymer-based hydrogels.^{5,13} Chemically and physically crosslinked double network hydrogels are well developed to improve the strength of hydrogels. However, chemical crosslinking methods usually leave unreacted chemical residues and cause poor biocompatibility.¹⁴ In addition, the chemically crosslinked structure is generally irreversible,¹⁵ which makes the hydrogel-forming process uncontrollable. Alternative strategies, such as micelle crosslinking, multiple hydrogen bonding, electrostatic interactions, and ionic crosslinking, have been developed to enhance the mechanical strengths of physically crosslinked hydrogels, but usually involve complex procedures and long reaction times.^{9,16} Achieving a balance between mechanical strength and applicable step-control preparation methods has become a challenge for the application of physical hydrogels in biological fields.

To solve these problems, a post-enhanced multi-network (PEMN) hydrogel was attempted to be constructed with natural polymers *via* a novel preforming dual-effect post-enhancing (PFDEPE) method. With the temperature-sensitive sol-gel transition reversibility of agar,^{17,18} a shape-controllable preforming hydrogel can be prepared by customized casting, injection, and 3D printing. Sodium alginate (SA) and chitosan (CTS) were selected due to their unzipped ionic crosslinks and stable electrostatic interactions to enhance the toughness.^{4,9} Based on the hierarchical chain entanglement developed by the PFDEPE method, the problems of uncontrollable preparation and low mechanical strength in traditional tissue engineering-used hydrogels are promised to be solved to meet the requirement of repairing irregular osteochondral defects. This work is conducive to exploring a novel solution and concept for preparing a kind of multiple all-physically cross-linked hydrogel based on sustainable natural polymers and their potential use in biological fields.

Results and discussion

Preparation and characterization of preformed hydrogels

In this work, the heating-mixing-cooling method was used to prepare preformed hydrogels with the desired mechanical strengths *via* a controllable formation process (Fig. 1). The contents and proportions of the raw materials for the hydrogel fabrication are shown in Table S1 (ESI[†]). The preformed hydrogels demonstrated thermal-reversible sol-gel transition behaviour and self-supporting abilities for the customized shape.

At a high temperature, a homogeneous and flowable liquid phase of agar solution enabled the uniform composites with other natural polymer powders. Therefore, hydrophilic SA polymer chains can dissolve completely in the neutral agar solution with high viscosity to support the dispersion of insoluble CTS and CaCO₃ powders and avoid phase separation.⁸ The composite

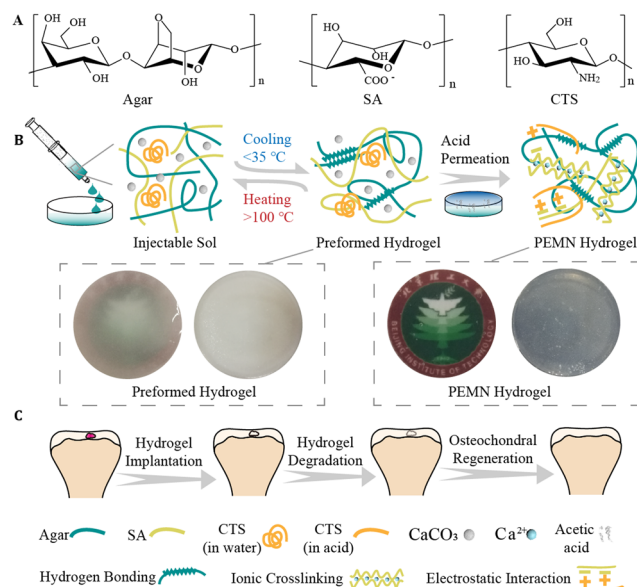


Fig. 1 PEMN hydrogel constructed by the PFDEPE method based on sustainable natural polymers. (A) Structures of Agar, SA, and CTS used to prepare the PEMN hydrogel. (B) Scheme of the PFDEPE method. (C) The application to efficient osteochondral regeneration.

system appeared as a fluid sol because the hydrogen bonds of agar were broken and no crosslinked network was formed between the components. As the temperature decreased, the complexed sol can undergo a rapid self-gelling process due to the helix structure transition of agar to form the first cross-linking network (Fig. 2(A)). The gelation time can be tuned by varying the polymer ratios (Table S2, ESI[†]). Owing to the temperature-sensitive properties, the sol can be cast into molds or extruded through syringes (Fig. 2(B) and Movie S1, ESI[†]). Meanwhile, 3D printing using the polymer sol as ink allows the fabrication of preformed hydrogels with the desired shapes (Fig. 2(C), (D) and Movie S2, ESI[†]).¹⁷ The preformed hydrogel with self-supporting properties can effectively fill and repair the

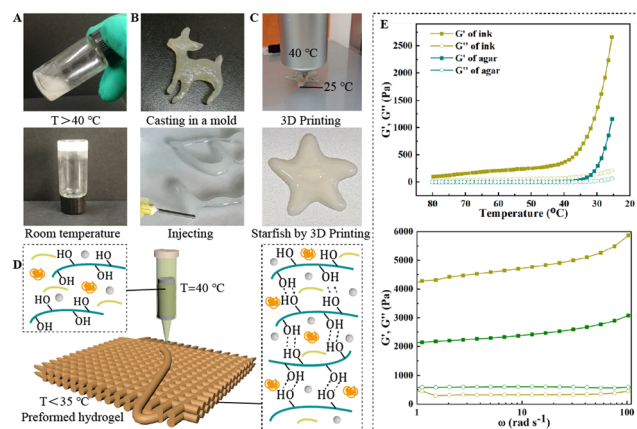


Fig. 2 The preparation of preformed hydrogel. (A) The transition of the polymer sol. (B) Casting and injection molding. (C) 3D printing molding. (D) The 3D printing principles. (E) Rheological properties of ink and agar solutions during the cooling process.

irregular-shaped defects at low natural polymer concentrations, and thus precisely customizes the appearance of osteochondral repair materials.^{3,19,20}

The rheological properties of the agar solution and polymer sol-based ink were investigated and compared. Both of them showed a temperature-sensitive behaviour in terms of storage modulus (G') and loss modulus (G'') (Fig. 2(E)). The sol-gel transition temperature of the ink during the cooling process was estimated to be 35–38 °C. Above the transition temperature, both G' and G'' of the ink were below 500 Pa, indicating its extrudability.²¹ As the temperature dropped below the phase transition temperature, G' and G'' increased sharply, suggesting a rapid enhancement in the mechanical strength of the preformed hydrogel, which enables rapid prototyping and morphological integrity after 3D printing without increasing the concentration of natural polymers.²² The temperature sensitivity arising from the physical hydrogen bonding within the agar allows the softened state of the preformed hydrogel to be restored by increasing the temperature. This controllable fabrication process offers convenience in adjusting the preparation steps and ingredients.

Preparation and characterization of PEMN hydrogels

The preformed hydrogel obtained above was then subjected to a dual-effect post-enhancing process to form a PEMN hydrogel. This novel PFDEPE method was inspired by the semi-dissolution acidification sol-gel transition (SD-A-SGT) method presented in our previous studies.^{8,14} In the SD-A-SGT method, a water-insoluble CTS powder was dispersed in a polyanion solution and exposed to acetic acid vapor, resulting in the formation of a homogeneous hydrogel *via* the electrostatic interaction between the positively charged $-\text{NH}_3^+$ on CTS chains and the negatively charged $-\text{COO}^-$ on the polyanion. However, the SD-A-SGT method requires a high concentration of acetic acid to generate the acid vapor, which is unsuitable for *in vivo* implantation. Moreover, excessive acid addition affects the hydrogen bonds in hydrogels, leading to a decrease in mechanical properties. To better explore the role of three physical networks in PEMN hydrogels, the Agar/SA/CTS (ASC) hydrogel and Agar/SA/ Ca^{2+} (ASCa) hydrogel prepared by the PFDEPE method, and the SA/CTS/ Ca^{2+} (SCCa) hydrogel and Agar/SA/CTS/ Ca^{2+} (ASCCa) hydrogel prepared by the SD-A-SGT method were obtained and characterized for comparison.

Benefiting from the preformed crosslinked network, the acid permeation can be applied to release Ca^{2+} and dissolve CTS for the formation of a PEMN hydrogel with a low concentration acetic acid solution (as low as 1%), instead of the saturated acetic acid used in SD-A-SGT. In the PFDEPE method, $-\text{NH}_2$ on CTS was protonated to the positively charged $-\text{NH}_3^+$, which interacts electrostatically with the negatively charged $-\text{COO}^-$ on the SA chain to form the second crosslinked electrolyte composite network. Meanwhile, the Ca^{2+} in CaCO_3 was dissociated and undergoes ion exchange with the Na^+ in the sodium carboxylate group on SA to form the third crosslinked network. The enhanced mechanical strength was achieved due to the combined effects of hydrogen bonding of agar, the ionic

crosslinking between Ca^{2+} and SA, and the electrostatic interaction between the protonated amino groups of CTS and the negatively charged carboxyl groups of SA. The homogeneous macro appearance and microstructures of the PEMN hydrogel are shown in Fig. S1 and S2 (ESI†) with uniform distribution of Ca (Ca^{2+}) and N (CTS) elements.²³

The structural differences between the preformed and PEMN hydrogels were characterized by XPS to explore the post-enhancing reaction mechanism.¹⁴ The broad-spectrum scan revealed that the Na peak in the preformed hydrogel disappeared in the PEMN hydrogel, indicating that the permeation of acetic acid causes the dissociation of Ca^{2+} from CaCO_3 and ion exchange with Na^+ from SA (Fig. S3, ESI†). High-resolution XPS spectra of each element are shown in Fig. 3(A). A new peak appeared in the N 1s XPS spectrum of the PEMN hydrogel assigned to the protonated $-\text{NH}_3^+$ (401.5 eV), indicating the protonation of $-\text{NH}_2$ on the CTS chain after acetic acid permeation. The high-resolution O 1s XPS spectrum of the preformed hydrogel was fitted into two peaks at 531.2 eV (C=O) and 532.8 eV (C–O), respectively. The binding energy of C=O became slightly increased in the PEMN hydrogel due to the coordination between $-\text{COO}^-$ and Ca^{2+} and the electrostatic interaction between $-\text{COO}^-$ and $-\text{NH}_3^+$. The high-resolution C 1s XPS spectrum of the preformed hydrogel was deconvoluted into three peaks, corresponding to O–C–O–C=O, C–O/C–N, and H–C/C–C, respectively. The peaks of

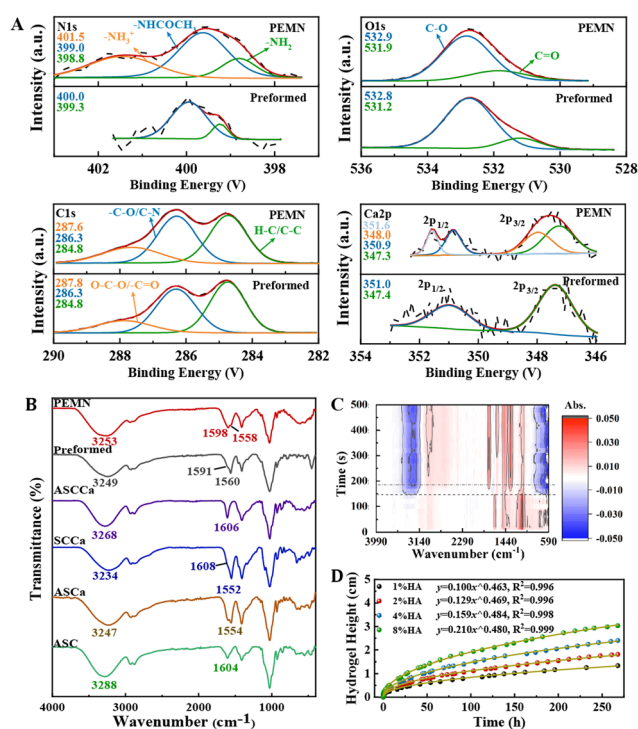


Fig. 3 PEMN hydrogel crosslinking structure characterizations. (A) XPS analysis of the preformed hydrogel and PEMN hydrogel. (B) ATR-FTIR results of ASC, ASCa, SCCa, ASCa, preformed, and PEMN hydrogels. (C) *In situ* ATR-FTIR results of the post-enhancing process. (D) Permeation kinetics and fitting of different acetic acid concentrations during the post-enhancing process.

O–C–O/–C=O and –C–O/C–N shifted to a lower binding energy region in the PEMN hydrogel, indicating the increased electron cloud density of the C atom due to interactions with Ca^{2+} , consistent with the change in the O 1s and N 1s spectra. The appearance of new Ca 2p_{1/2} and Ca 2p_{3/2} peaks in the PEMN hydrogel confirmed the occurrence of the coordination between Ca^{2+} and the –COO[–] on SA chains.

The ATR-FTIR spectra of the hydrogels are shown in Fig. 3(B). The absorption peaks of ASC and SCCa hydrogels at 1604 cm^{–1} and 1608 cm^{–1} were assigned to the stretching vibration of N–H in –NH₂ and –NH₃⁺, respectively, indicating the protonation of the amino group of CTS. In addition, the C=O peak in SA blue shifted (from 1593 to 1554 cm^{–1} and 1552 cm^{–1}, respectively) in ASCa and SCCa hydrogels,²⁴ suggesting the occurrence of the coordination between –COO[–] and Ca^{2+} . Similarly, the absorption peaks of PEMN and ASCCa hydrogels at 1598 cm^{–1} and 1610 cm^{–1} were assigned to the stretching vibration of N–H in –NH₃⁺.⁸ Due to the lower concentration of acid used in the PFDEPE method compared to the SD-A-SGT method, the hydrogen bonds in the PEMN hydrogel were less disrupted,⁷ as evidenced by the lower wavenumber of the O–H peak at 3253 cm^{–1} in the PEMN hydrogel and 3268 cm^{–1} in the ASCCa hydrogel. The comparison of FTIR spectra of different hydrogels revealed the formation of multiple physically crosslinked networks and stronger hydrogen bonds in the PEMN hydrogel.

Permeation kinetics

To further understand the reactions occurring during the acidification dual-effect post-enhancing process, *in situ* real-time ATR-FTIR was conducted (Fig. 3(C)). The absorption peaks at 1711 cm^{–1} were clearly observed when 1% acetic acid solution was added after 150 s, while the absorption peaks of water at 3400^{–1} became significantly weaker, indicating the crosslinking of the multiple physical networks was a drainage process and the permeation equilibrium was achieved as the crosslinked structure was completely formed. Meanwhile, an absorption peak gradually appeared at 1608 cm^{–1}, which can be attributed to the conversion of –NH₂ on chitosan to –NH₃⁺. These results suggest that the acetic acid solution effectively infiltrated the preformed hydrogel to acidify CTS and CaCO₃ and achieve the permeation equilibrium inside and outside the hydrogel.

The acidification dual-effect post-enhancing process to form PEMN hydrogel is also controllable. The transparency change of the hydrogel serves as a visual indicator to demonstrate the gradual dissolution of CTS and CaCO₃ during the post-enhancing process and the controllability of the preparation (Fig. S4, ESI†). Therefore, the kinetics of the post-enhancing process can be determined by monitoring the transparency of the preformed hydrogels acidified with different concentrations of acetic acid. As shown in Fig. 3(D), the kinetics of the post-enhancing process to fabricate PEMN hydrogel was a power function. With increasing acetic acid concentration from 1% to 8%, the formation time of a 1 cm PEMN hydrogel was reduced to 17.9% of the original time. This indicates that the

formation time of PEMN hydrogel can be tuned and controlled by adjusting the concentration of acetic acid.

Mechanical and morphological characterization

Mechanical strength is an important parameter affecting the supporting properties of an implant material for articular osteochondral. The multiple physically crosslinked networks in PEMN hydrogel make it possible to overcome the poor mechanical properties of natural polymer-based hydrogels in osteochondral repair.^{22,23} The mechanical test results of the PEMN hydrogels prepared with different polymer contents and comparison hydrogels are shown in Fig. 4(A) and (B).

The tensile strength of the PEMN hydrogel with 2.7% polymer contents prepared using the PFDEPE method was higher (0.21 MPa) than those of the ASC hydrogel (0.03 MPa), ASCa hydrogel (0.07 MPa), SCCa hydrogel (0.14 MPa), and ASCCa hydrogel (0.18 MPa). The tensile strength and elastic modulus of the PEMN hydrogel improved with the increase of polymer contents in the same ratio owing to the larger number of crosslinking sites.²⁵ For example, the tensile strength (0.51 MPa) and elastic modulus (0.64 MPa) of the PEMN-6.3% were higher than those of PEMN-2.7%. Also, the compressive strength and elastic modulus of the PEMN hydrogel can be tuned in the ranges of 0.34–1.26 MPa and 0.16–0.64 MPa respectively by adjusting the polymer contents of the hydrogel,

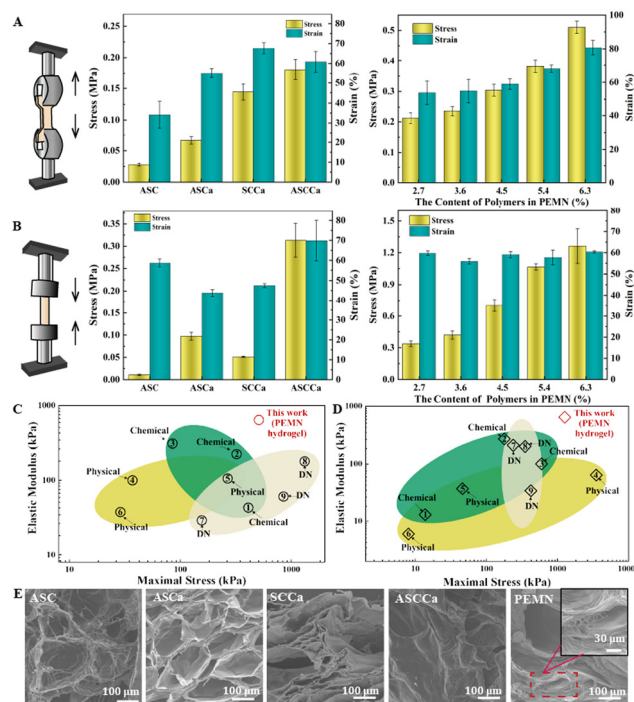


Fig. 4 Mechanical behaviour and morphology characterizations. (A) Tensile strength of hydrogels. (B) Compressive strength of hydrogels. (C) Comparison of the maximum tensile stress and elastic modulus with reference. (D) Comparison of the maximum compressive stress and elastic modulus with reference. No. 1–3 indicate chemical hydrogels for tensile^{19,29,30} and compression.^{31–33} No. 4–6 indicate physical hydrogels for tensile^{8,35,36} and compression.^{28,37,38} No. 7–9 indicate physical/chemical double network hydrogels for tensile^{9,39,40} and compression.^{27,41,42} (E) SEM morphology of hydrogels.

which meets the requirement of 0.5 MPa for knee cartilage (Table S3, ESI†).²⁶ The mechanical strength at 37 °C was slightly lower due to the disruption of some hydrogen bonds but still remained within practical use requirements. The fracture energy of PEMN hydrogels ranged from 493.63–2384.67 J m⁻², which was higher than that of articular cartilage (~1000 J m⁻²).^{9,12,27} The cyclic tensile test demonstrated that even after 50 cycles, PEMN hydrogel retained over 75% of its initial tensile strength, indicating its potential for long-term use in osteochondral applications (Fig. S5, ESI†).²⁸ Owing to the energy-dissipative strong electrostatic interactions, ionic crosslinks, and hydrogen bonds in PEMN hydrogel, excellent mechanical properties were achieved at the water content of 85%, which is similar to that of cartilage tissue (Fig. S6, ESI†).⁹ Although the PEMN hydrogel is completely crosslinked by physical interactions, it exhibited superior mechanical properties compared to covalently crosslinked chemical hydrogels,^{19,29–34} non-covalently crosslinked physical hydrogels,^{8,28,35–38} and physical/chemical double network hydrogels,^{9,27,39–42} particularly in terms of tensile and compression elastic modulus (Fig. 4(C) and (D)).

The rheological properties of PEMN hydrogels were further characterized to understand their microstructural crosslinking.¹⁸ In the whole scan range, the G' of the PEMN hydrogel was higher than G'' , showing the typical rheological behaviours of the hydrogel (Fig. S7, ESI†).²³ Continuous step-strain measurements were also conducted to investigate the rheological recovery behaviours of the PEMN hydrogel. The hydrogel was subjected to small (1%) and large (200%) oscillatory shear strains alternately. At the 200% strain stage, G'' was much larger than G' , indicating damage to the PEMN hydrogel network. Upon restoring the strain to 1%, both G' and G'' immediately returned to their original values, suggesting the excellent dynamic reversibility of the multiple physical networks in the hydrogel.⁴³

The swelling properties reflect the stability of crosslinked networks. The swelling ratios of the PEMN hydrogels and comparison hydrogels were measured by the 24 h equilibrium swelling test. The low swelling ratios of the PEMN hydrogels ranging from 6–14 g g⁻¹ (Fig. S8, ESI†) were possibly attributed to the high density of multiple physically crosslinked networks, which restricts the diffusion of water molecules into the hydrogel, consistent with rheological and mechanical results. The low swelling ratio ensures the hydrogel volume remains relatively stable during long-term use and subsequent implantation.^{8,44}

The microscopic morphologies of different hydrogels were characterized to understand the strong mechanical strength of the PEMN hydrogel. In Fig. 4(E), the microstructure of the ASC hydrogel exhibited square pore-like structures composed of filamentous links. The ASCa hydrogel displayed a rhombic pore-like structure, with smaller pore sizes compared to the ASC hydrogel. The SCCa hydrogel exhibited a lamellar structure and the ASCCa hydrogel was compact and dense. The microstructure of the PEMN hydrogel showed a combination of porous and lamellar structures, indicating it contained different types of crosslinked networks. In addition, filamentous structures were prominently observed in the lamellar and pore structures of the PEMN hydrogel, which increases the mechanical strength and fracture

toughness of the PEMN hydrogel due to a mechanical dissipation effect to meet the requirements of osteochondral repair.^{9,14,39,42}

In general, the mechanical enhancement of the PENN hydrogel can be ascribed to the hierarchical chain entanglement achieved through the PFDEPE method. The preformed hydrogel crosslinked by hydrogen bonds acts as a porous and continuous supporting layer to hold the individual polymer chains. The electrostatic interaction and ionic crosslinking presented as the filamentous structure growing within the supporting layer further enhance the bonding between polymer chains and provide mechanical dissipation effects. The hierarchical chain entanglement formed by the supporting layer and the filamentous structure increases the number of crosslinking points and ensures the toughness of the PEMN hydrogel. Besides, the endpoint of the reaction is easily controlled by the PFDEPE method, which avoids the excessive acid entering the hydrogel network to break the hydrogen bonding of the supporting layer.¹⁴ These factors contribute to the enhanced mechanical properties for further osteochondral defect repair application.

Adhesion performance

In addition to the excellent mechanical properties, the PEMN hydrogel also exhibited good adhesive properties, which make it applicable for irregular defects along with its self-supporting ability. The defect repair test was performed on a chipped paper cup. A circular notch with a diameter of ~30 mm was created, and the polymer sol was extruded into the notch with a syringe. Due to the self-supporting effect, the sol was rapidly cooled and crosslinked to form a preformed hydrogel that filled the entire defect (Fig. 5(A) and Movie S3, ESI†). The preformed hydrogel was then subjected to the acidification dual-effect

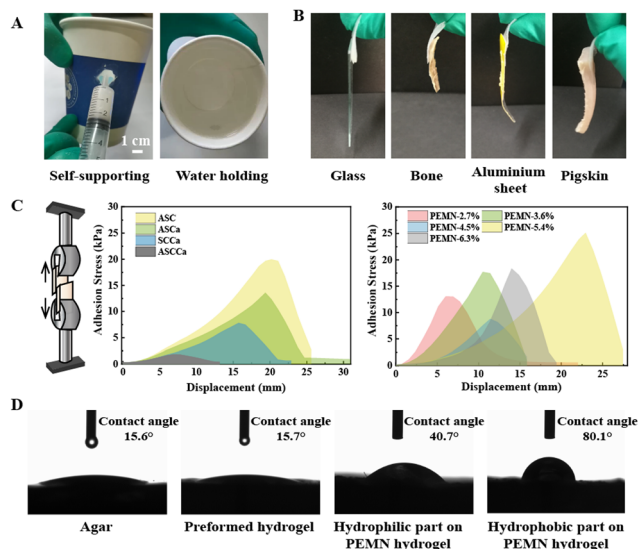


Fig. 5 Self-supporting and adhesive properties of hydrogels. (A) Self-supporting property of PEMN hydrogels to hold deionized water for more than 7 days. (B) The adherence of PEMN hydrogel to different solid surfaces. (C) The adhesion curves of the hydrogels. (D) Contact angle test of agar, preformed hydrogel, and PEMN hydrogel.

post-enhancing process to eventually form the PEMN hydrogel with certain adhesion abilities. It sturdily adhered to the paper cup surface, and the repaired cup was able to hold water without leakage for over a week at room temperature.

PEMN hydrogels can adhere to both polar and non-polar surfaces, including glass, animal bones, aluminium sheets, and pigskin (Fig. 5(B)). Adhesion behaviours of the hydrogel that underwent a sol-gel transaction and acidification double-effect post-enhancing process *in situ* on pigskin were evaluated by lap shear tensile tests. PEMN-5.4% hydrogel exhibited the highest adhesive strength of 25.14 kPa, comparable to that of commercial fibrin glue (13.2 kPa). The adhesion energy of PEMN-5.4% was 249.66 J m^{-2} (Fig. 5(C)), which is approximately 4.8 times greater than that of commercial fibrin glue (52.3 J m^{-2}).²⁵ A practical demonstration of the adhesion properties was observed when the PEMN hydrogel was applied between two pieces of pigskin, successfully supporting a weight of 50 g (Movie S4, ESI†). The results of the bioadhesion between hydrogels directly adhered to the pigskin are presented in Fig. S9 (ESI†), demonstrating the possibility of PEMN hydrogel adhering to the defect site as a scaffold *in vivo*.

The adhesion properties of PEMN hydrogel can be explained for two main reasons. Firstly, multiple hydroxyl groups in the hydrogels facilitate the formation of hydrogen bonds with various surfaces, thereby improving the cohesive force and the adhesion performance together with ionic interactions. Secondly, the cross-linked networks reduce the hydrophilicity, leading to enhanced interaction between the PEMN hydrogel and the substrate surface. The contact angles of both agar and preformed hydrogel were $\sim 15.6^\circ$, while the contact angle of PEMN hydrogel increased to 40.7° for the more hydrophilic part from hydrophilic natural polymers, and 80.1° for the more hydrophobic part from the 3D crosslinking network (Fig. 5(D)), which contributes to the good adhesion performance of the hydrogel by preventing water molecules from penetrating into the network, thus inhibiting volume changes and minimizing the expansion of the defect site.⁴⁴

Hydrogel degradation *in vitro*

To assess the potential of the PEMN hydrogel as a tissue engineering scaffold, its degradation under physiological conditions was investigated.³ The ASC hydrogel was selected for comparison to analyse the effects of Ca^{2+} and the mechanical behaviours. The PEMN and ASC hydrogels were immersed in phosphate-balanced saline (PBS), PBS with $2 \mu\text{g mL}^{-1}$ collagenase II, and PBS with $10 \mu\text{g mL}^{-1}$ lysozyme for 12 weeks, respectively. As shown in Fig. S10 (ESI†), the degradation of PEMN can be demonstrated by the ATR-FTIR spectrum and gel permeation chromatography (GPC). The degradation behaviours of PEMN and ASC hydrogels in PBS and PBS with lysozyme were similar because the crosslinking of CTS and SA reduced the degradation performance of lysozyme on CTS. In the collagenase II solution, ASC hydrogel underwent rapid degradation, while the degradation of PEMN hydrogel was slower, possibly due to the dense 3D structures of the strong ionic coordination between SA and Ca^{2+} . The degradation process of the hydrogel may involve two main stages:⁴⁵ first, the break of the post-enhancement network, and

second, the thermally induced collapse of the agar matrix. The SEM images of the hydrogels degraded for 12 weeks revealed collapse of the pores, and destruction of the 3D structure of the hydrogel after degradation, which is beneficial for the growth of newly generated osteochondral tissue.⁴⁶

Effects of cell viability and differentiation *in vitro*

The cell proliferation was evaluated by incubating rat bone marrow mesenchymal stem cells (rBMSCs) on different hydrogels (Fig. 6(A) and (B)).⁴⁷ The cell viability of the PEMN hydrogel increased up to 164%, indicating promoting effects of the hydrogels on cell proliferation, while the cell viability in ASC, ASCa, SCCa, and ASCCa hydrogels was 106%, 58%, 64.1%, and 61.3%, respectively. It can be explained that proliferation requires both a certain stiffness and biocompatibility. For ASC and ASCa hydrogels, insufficient mechanical strength support limited the 3D spreading of rBMSCs.⁴³ The lack of positively charged CTS in the ASCa hydrogel further reduced cell adhesion. For SCCa and ASCCa hydrogels prepared by the SD-A-SGT method, acetic acid residues were difficult to remove and detrimental to the growth of rBMSCs, while the acidification reaction of the PEMN hydrogel was easily monitored by observing the transparency change, thereby preventing excessive acid from entering the hydrogel network.⁴⁸ This was confirmed in the biocompatibility tests with hydrogel extracts (Fig. S11, ESI†). Therefore, the suitable elastic modulus and mild preparation conditions of the PEMN hydrogel made it suitable for cell growth.

The promoting effects of the hydrogels on osteogenic effect were further investigated by alizarin red staining. rBMSCs were cultured in the presence of PEMN hydrogel and ASC hydrogel and in the absence of hydrogel (blank control), respectively (Fig. 6(C)). Both the PEMN hydrogel group and ASC hydrogel group showed stronger osteogenic-induced differentiation effects than the control group after 7 and 14 days of culture.²⁵ The PEMN hydrogel group demonstrated greater calcium deposition than the other two groups, indicating its ability to promote osteogenic

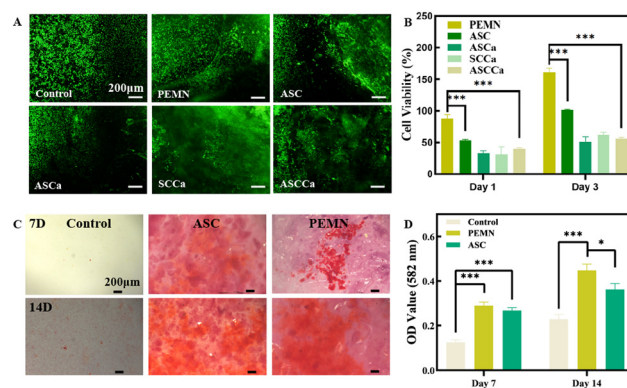


Fig. 6 Cell proliferation activity and differentiation on hydrogels. (A) Live/dead staining of rBMSCs for 24 h. (B) The proliferation of rBMSCs cultured on the surface of different hydrogels. (C) Alizarin red staining with different hydrogels for day 7 and day 14. (D) quantitative analysis of the alizarin red. * denotes $p < 0.05$ and *** denotes $p < 0.001$ compared with the control group.

differentiation of rBMSCs.²⁰ Quantitative analysis further confirmed more differentiated cells in the PEMN hydrogel and ASC hydrogel groups than in the blank control group on day 7 (Fig. 6(D)). Similarly, after 14 days of culture, the PEMN hydrogel group exhibited significantly higher alkaline phosphatase (ALP) expression than those of the two groups (Fig. S12, ESI†). These cytocompatibility and induced differentiation analyses showed that PEMN hydrogel has the potential to induce osteogenic differentiation by long-term co-incubation.

Osteochondral regeneration *in vivo*

An *in vivo* osteochondral repair experiment with a control group, PEMN hydrogel group, and ASC hydrogel group was conducted using the osteochondral defect model.³ Defects of 2 mm in diameter and 3 mm in depth were created, and the hydrogels were implanted in the defects of the experimental groups respectively.

The biocompatibility of the hydrogels after implantation *in vivo* was evaluated by H&E staining of the kidney and liver and blood biochemical tests (Fig. S13, ESI†). No significant differences were seen between the hydrogel implantation group and normal rats, indicating few infections or implant rejections.

The osteochondral repair performance of each hydrogel was evaluated by Micro-CT at week 6 and week 12 (Fig. 7(A)). The 3D reconstruction and 2D images of the osteochondral defects taken at week 6 revealed more new bone formed in the PEMN hydrogel and ASC hydrogel groups compared to the control group with untreated defects. By week 12, the defects in the control group were less filled with the newly formed cartilage and subchondral bone, indicating poor osteochondral repair ability. The PEMN hydrogel group exhibited more complete regeneration of cartilage and subchondral bone in the implanted defect, with a low center defect possibly due to the mechanical supporting ability.⁴⁸ The ratio of bone volume to tissue volume (BV/TV) and bone mineral density (BMD) of the regenerated bones were determined by quantitative micro-CT analysis (Fig. 7(B)). The BV/TV of the PEMN hydrogel group was 28.82% at week 6, significantly higher than that of the blank group (7.98%) and ASC hydrogel group (17.22%). The cartilage and subchondral bone were further repaired by week 12 with BV/TV increased to 15.8%, 44.3%, and 32.6% for the blank, PEMN hydrogel, and ASC hydrogel groups, respectively.

The results of Micro-CT showed that the new bone tissue initially formed on the surface of the cylindrical PEMN hydrogel, and the central vacancies were filled with time. It can be explained that the lateral hydrogel was more susceptible to the influence of the internal microenvironment and enzymes, making the iteration between hydrogel degradation and tissue regeneration more likely to occur. The excellent mechanical strength, adhesion, and biocompatibility of the PEMN hydrogel provided mechanical support to the regenerated bone and cartilage and prevented further damage to enlarge the defect during the movement.³ The supporting structure of hydrogel with a large surface area can host more osteogenic cells and promote tissue regeneration. In addition, the SA and Ca²⁺ crosslinked network in the PEMN hydrogel provides Ca²⁺ that

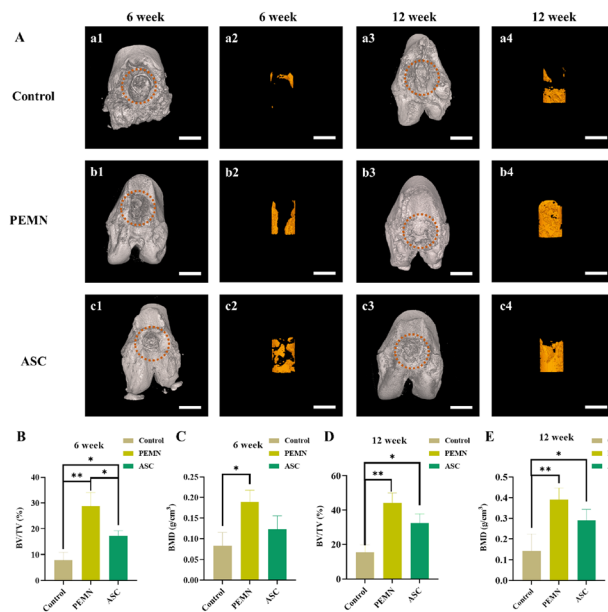


Fig. 7 Regeneration of cartilage-subchondral bone in critical-sized defects at week 6 and 12 after implantation. (A) 2D Micro-CT images, including scanning images (a1–c1, a3–c3) and Micro CT reconstruction images (a2–c2, a4–c4) (scale bars = 2 mm). (B and D) Micro-CT bone volume fraction. (C and E) bone mineral density values. * denotes $p < 0.05$ and ** denotes $p < 0.01$ compared with the control group.

promotes endogenous cell mineralization and enhances the bone viability within the PEMN hydrogel for the construction of new bone tissues closely connected with the host.⁴⁷ The tissue regeneration in the osteochondral defect of PEMN hydrogels was completed by the late repair stage at week 12, demonstrating the strong promoting effects of the hydrogel on tissue regeneration.

Histopathology analysis

The osteochondral regeneration of PEMN and ASC hydrogels were also evaluated by histological and immunohistochemical (IHC) staining using H&E, Masson, SAF-O/Fast Green, and Col II at week 6 and 12 postoperatively (Fig. 8).

The H&E and Masson staining results showed no obvious inflammation around the implanted hydrogel. The tissues repaired by the PEMN hydrogel at week 12 exhibited uniformly arranged chondrocytes, a large number of spindle fibroblasts, and well-organized regenerated bone, indicating the PEMN hydrogel can simultaneously enhance the repair of articular cartilage and subchondral bone.^{20,47} Due to the lack of mechanical support and the osteoinductive effect of Ca²⁺, there was much less new bone in the subchondral bone region in the ASC hydrogel group. In contrast, few fibrocytes in the defect of the control group with untreated defect were observed even at week 12. These results indicated that a large number of cations and the stimulation of Ca²⁺ ions in the PEMN hydrogel accumulated large amounts of osteoinductive proteins and active factors in the defect site, which further promoted cell recruitment, migration and differentiation, and new bone formation.⁴⁶ Meanwhile, the formation of cartilage occurred during intrinsic osteoinduction.⁴⁹

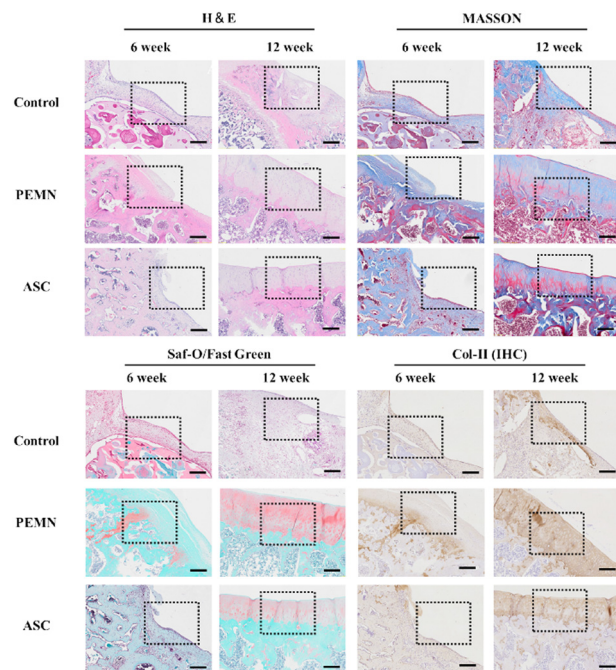


Fig. 8 Pathological analysis in week 6 and 12 with black dotted boxes marked as areas of osteochondral defect, including H&E, Masson, SaF-O/Fast Green, and Col II staining. (Scale bars = 250 μm).

The cartilage repair was also evaluated by the SaF-O/Fast Green and Col II staining. For the control group with untreated defects, only a thin layer of cartilage tissue appeared on the articular surface and obvious vacancies in the defect sites were observed at week 12 with a weakly positive SaF-O/Fast Green and Col II staining signal due to the limited self-repair ability. In the ASC hydrogel group, there was a slightly larger positive area for Col II staining at week 12 compared to the control group. For the PEMN hydrogel group, a large number of regularly arranged Col II fibers were observed in the stained section, indicating high Col II expression with a strongly positive staining signal, even reaching that of natural cartilage.¹⁹ The SaF-O/Fast Green staining demonstrated that the regenerated hyaline chondrocytes in the PEMN hydrogel group were arranged parallel to the surface in the superficial region and the chondrocytes in the cartilage layer were

columnar. The mechanical supporting structure and cell recruitment effects of the PEMN hydrogel facilitated chondrocyte differentiation and extracellular matrix deposition.^{3,13} Thus, a smooth cartilage repair layer was formed and thickened over time. Eventually, the boundary between the repaired cartilage and host bone was indistinguishable. Calcified bone tissues were found below the cartilage and the mature trabecular bone grew inwards. These results showed the PEMN hydrogel can promote the simultaneous regeneration of both cartilage and subchondral bone.

In cartilage, bone, and osteochondral tissue engineering, the importance of mechanical integrity and customizable shape has been recognized as significant research directions.^{2,6} To solve this problem, previous studies proposed the injection-moldable method to enable the adaptivity of hydrogels to irregular defects,¹³ as listed in Table 1. However, most of the existing systems adopted chemical cross-linking networks,^{19,50–52} which may fail to meet the mechanical requirements of cartilage and bone and leave toxic crosslinking reagents.²⁴ In this study, a PEMN hydrogel with hierarchical chain entanglement was developed to achieve a controllable shape and mechanical support. Compared to previous studies, the improved BV/TV and BMD value showed osteochondral regeneration promotion with PEMN hydrogel implantation.

Conclusions

In this work, the PEMN hydrogel based on natural polymers with hierarchical chain entanglement was prepared by a novel PFDEPE method to address the challenge of achieving both mechanical strength and controllable preparation. The pre-forming process can precisely customize the hydrogel shape to fit irregular defects. The hierarchical chain entanglement of the multiple physically crosslinked networks in the hydrogel resulted in impressive mechanical properties of 1.26 MPa at a water content of 85% and a fracture energy of 2384.67 J m⁻², meeting the mechanical requirements for osteochondral repair materials and ensuring shape stability in the defect site after being implanted. The mechanical support property, adhesiveness, biocompatibility, degradability, and Ca²⁺ of the PEMN hydrogel make it a great scaffold candidate for osteochondral

Table 1 Comparison between the QCM humidity sensor in this work and reported works

Materials	Mechanical strength	Customized shape	Degradation	BV/TV (%)	BMD (g cm ⁻³)
Methacrylated sodium alginate, gelatin methacryloyl, and β -tricalcium phosphate ⁵⁰	0.24 MPa for compressive stress	Yes	More than 4 weeks	~35	—
SA/AG + ACs/BMSCs ⁵¹	19 kPa for compressive modulus	Yes	—	39.75	0.22
CSMA-NIPAm/PECDA-AAm-PEGDA ⁵²	~0.3 MPa and ~0.154 MPa for compressive stress and modulus	No	More than 8 weeks	~65	—
N-Acryloyl glycineamide, and N-[tris(hydroxymethyl) methyl] acrylamide ¹⁹	0.41 MPa for tensile stress	Yes	—	~33	~0.14
Polyetheretherketone/tilted nanolamellae ¹³	—	No	—	~48	—
SA, gellan gum, and thixotropic magnesium phosphate-based gel ²⁴	811 kPa for compressive stress	Yes	—	35.27	—
PEMN hydrogel (this work)	0.34–1.26 and 0.16–0.64 MPa for compressive stress and modulus	Yes	More than 12 weeks	44.8	0.39

regeneration, especially for endogenous cell mineralization and chondrocyte differentiation. In general, the novel PFDEPE method for sustainable natural polymer-based physical hydrogels with precisely customizable shapes and high toughness shows outstanding translational prospects in the clinical application of osteochondral regeneration.

Author contributions

Conceptualization: J. Yang, Y. Chen; methodology: J. Yang, H. Wang, W. Huang, K. Peng, L. Lin, J. Yuan, W. Yao, X. Ma; investigation: J. Yang, H. Wang, W. Huang; supervision: R. Shi, W. Tian, Y. Chen; writing – original draft: J. Yang, H. Wang; writing – review and editing: K. Lin, Y. Chen.

Conflicts of interest

There are no conflicts to declare.

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